

Exploring a Running z/OS System via SDSF





SYSTEM DISPLAY AND SEARCH FACILITY

You've used the System Display and Search Facility (SDSF) to check the status of jobs and to view the spooler output from jobs. SDSF also displays information about the z/OS instance, configuration details, current operational state, and subsystems.

As a Platform Engineer or z/OS System Programmer your userid should be authorized to access most or all these displays.

In this module, we're going to explore the z/OS system we're using for lab work and discover what kinds of information we can access with SDSF.

In the process, we'll also strengthen our understanding of what happens in a live z/OS environment. We need that knowledge when we're tasked with performing system administration work, whether on platform or via an external tool like Ansible.

SDSF MENU 3.1		LOCAL	VS01	
COMMAND INPUT		→		
NP	NAME	Description	Group	Status
	JG	Job groups	JES	
	SYM	System symbols	System	
	LOG	System log	Log	
	SR	System requests	Log	
	MAS	Members in the MAS	JES	
	JC	Job classes	JES	
	SE	Scheduling environments	WLM	
	RES	WLM resources	WLM	
	ENC	Enclaves	WLM	
	PS	Processes	OMVS	
	SYS	System information	System	
	ENQ	Enqueues	System	
	ENQC	Enqueue contention	System	
	ENQD	Enqueued data sets	Sysplex	
	DYNX	Dynamic exits	System	
	AS	Address space memory	Jobs	
	INIT	Initiators	JES	
	PR	Printers	JES	
	PUN	Punches	JES	
	RDR	Readers	JES	
	LINE	Lines	Network	
	NODE	Nodes	Network	
	SO	Spool offload	JES	
	SP	Spool volumes	JES	
	NS	Network servers	Network	
	NC	Network connections	Network	
	RM	Resource monitor	JES	
	CK	Health checker	System	
	LNK	Link list data sets	System	
	LPA	Link pack data sets	System	

SYSTEM DISPLAY AND SEARCH FACILITY

```
SDSF MENU 3.1    LOCAL    VS01
COMMAND INPUT ==>
```

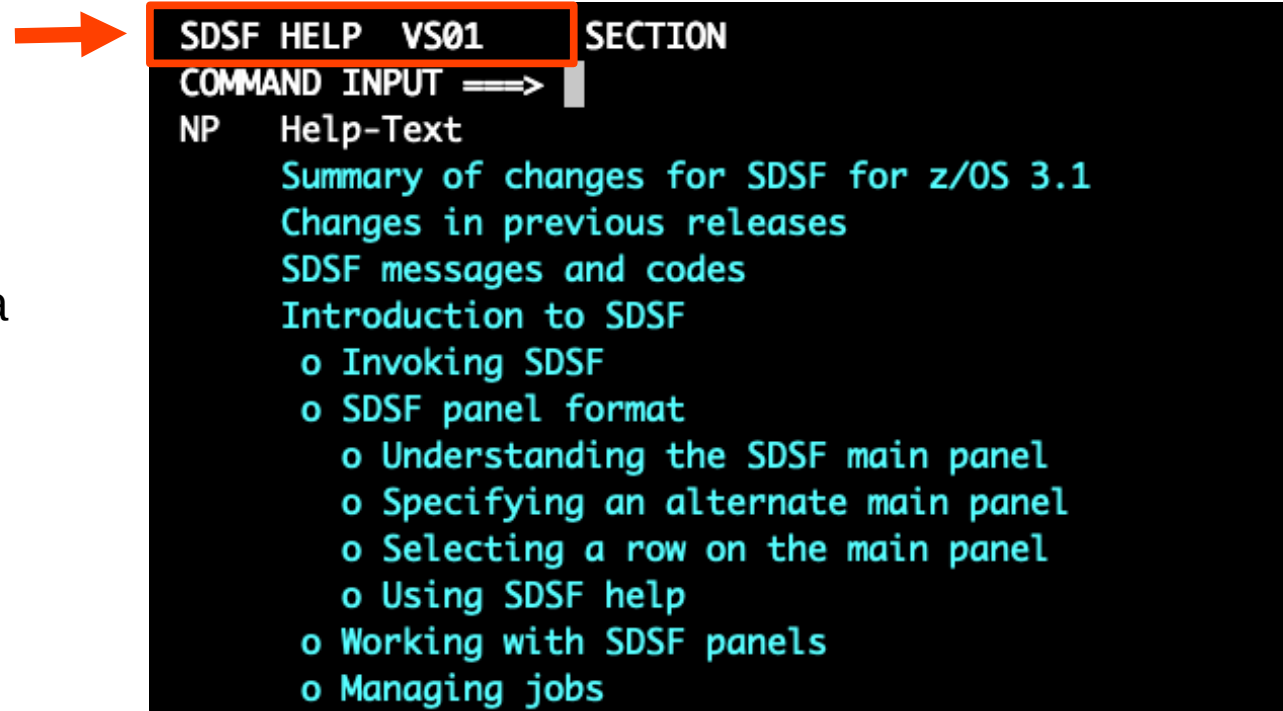
NP	NAME	Description	Group	Status
	EDT	Eligible device table	Devices	
	PPT	Program properties	Program	
	JRG	JES resource groups	JES	
	JRJC	JES class resource limit	JES	
	SMFO	SMF options	System	
	SMFD	SMF data sets	System	
	SMFS	SMF subsystems	System	
	DASH	Dashboard	System	
	ELOG	Event log	System	
	ULOG	User session log	Log	
s	HELP	SDSF help facility	SDSF	

Let's start by scrolling all the way to the bottom of the list, to HELP. This is where to go when you're not sure what something means or where to find information you're looking for.



SDSF HELP

The HELP panel tells us we're looking at help for system VS01. The topics are organized as a table of contents. Select a topic by typing an "S" next to the title and pressing Enter.



```
SDSF HELP VS01 SECTION
COMMAND INPUT ==>
NP  Help-Text
    Summary of changes for SDSF for z/OS 3.1
    Changes in previous releases
    SDSF messages and codes
    Introduction to SDSF
      o Invoking SDSF
      o SDSF panel format
        o Understanding the SDSF main panel
        o Specifying an alternate main panel
        o Selecting a row on the main panel
        o Using SDSF help
      o Working with SDSF panels
      o Managing jobs
```

SDSF HELP FOR THE SYSTEM PANEL

```
SDSF HELP  VS01      SECTION
COMMAND INPUT ==>
NP  Help-Text
s  o System panel (SYS)
   o System Parameters panel (SYSP)
   o System Requests panel (SR)
   o Unit Control Blocks panel (UCB)
   o Virtual Storage Map panel (VMAP)
   o WLM Policy panel (WLM)
   o WLM Report Class panel (REPC)
   o WLM Resource Group panel (RGRP)
   o WLM Service Classes panel (SRVC)
   o WLM Workload panel (WKLD)
   o XCF Application Servers panel (XCFA)
   o XCF Members and Groups panel (XCFM)
   o XCF Signaling Paths panel (XCFP)
```



We'd like to learn how to find information about the system.

We scroll down until we see "System panel" and select that entry.

SDSF HELP FOR THE SYSTEM PANEL

This provides detailed information about everything that is displayed on the System panel. If you're unsure about what anything means in SDSF, you can find an explanation under HELP.

```
SDSF HELP VS01      SECTION SYS          LINE 1-36 (349)
COMMAND INPUT ==> █                                SCROLL ==> CSR
NP   Help-Text
     System panel (SYS)
```

The System Panel (SYS) allows you to display information about systems in the sysplex such as CPU busy, storage utilization, and IPL information.

Command

Access the System panel with the SYS command from any SDSF panel.

Panel title information

The title line contains the following information:

SDSF SYSTEMS	SYS1	SYS*	PLEXNAME	LINE 1-1 (1)
				Total lines
System you are logged on to			Sysplex name	
	Systems displayed (MVS value or SYSNAME value)			Lines displayed

SORTING THE ENTRIES ON THE MAIN PANEL

SDSF MENU 3.1 LOCAL VS01				
COMMAND INPUT ==> sort group				
NP	NAME	Description	Group	Status
	AD	Address space diagnostic	Jobs	
	APF	APF data sets	System	
	AS	Address space memory	Jobs	
	BPX0	OMVS options	OMVS	
	CF	Coupling facilities	Sysplex	
	CFC	CF connections	Sysplex	
	CFD	Couple data sets	Sysplex	
	CFS	CF structures	Sysplex	
	CFSA	CF structure activity	Sysplex	
	CK	Health checker	System	
	CS	Common storage subpools	Memory	
	CSR	Common storage remaining	Memory	



We exited from HELP by pressing F3 a couple of times, and now we want to explore System information.

We can re-order the entries on the main SDSF panel with the SORT command. Enter SORT <columnname> to rearrange the entries.

We're interested in System information, so we'll enter SORT GROUP and then find the System entries.

SDSF TOPICS UNDER THE SYSTEM GROUP

SDSF can display quite a lot of information about the z/OS instance. Let's begin with SYS, "System information."



SDSF MENU 3.1		LOCAL	VS01	
COMMAND INPUT		⇒		
NP	NAME	Description	Group	Status
	SYM	System symbols	System	
	SYS	System information	System	
	ENQ	Enqueues	System	
	ENQC	Enqueue contention	System	
	DYNX	Dynamic exits	System	
	CK	Health checker	System	
	LNK	Link list data sets	System	
	LPA	Link pack data sets	System	
	APF	APF data sets	System	
	PAG	Page data sets	System	
	PARM	Parmlib data sets	System	
	SSI	Subsystem information	System	
	GT	Generic tracker	System	
	LLS	Link list sets	System	
	SYSP	System parameters	System	
	LPAR	Logical partitions	System	
	SMFO	SMF options	System	
	SMFD	SMF data sets	System	
	SMFS	SMF subsystems	System	
	DASH	Dashboard	System	
	ELOG	Event log	System	

SYS – SYSTEM INFORMATION

SDSF SYSTEMS		VS01	VS01	LOCAL		LINE 1-1 (1)					
COMMAND INPUT		=>		SCROLL => CSR							
NP	SYSNAME	SysLevel	CPU%	SIO	Aux%	CSA%	SQA%	ECSA%	ESQA%	UIC	Spool%
	VS01	z/OS 03.01	9	0	0	8	13	9	10	65535	0

scrolled right once

SDSF SYSTEMS		VS01	VS01	LOCAL		LINE 1-1 (1)					
COMMAND INPUT		=>		SCROLL => CSR							
NP	SYSNAME	CADS%	PageRate	Real	RealAFC	RealAFCB	Fix%	FixB%	MaxASID	Free	
	VS01	38	0	3607909	2328934	3849	1	0	2045		

scrolled right twice

SDSF SYSTEMS		VS01	VS01	LOCAL		LINE 1-1 (1)					
COMMAND INPUT		⇒		SCROLL ⇒ CSR							
NP	SYSNAME	ASID	BadASID	STC	TSU	Job	WTOR	Sysplex	LPAR	VMUser	
	VS01	1947	2	78	1	0	1	LOCAL	VS01		

SDSF displays the basic System information as a single line. We have to scroll to the right to see all the fields.

We can see, for instance, that we're running z/OS version 3.1, the CPU is currently 9% utilized, the Extended System Queue Area is 10% full, and our Address Space ID is 1947.

SDSF SYSTEM DASHBOARD

The DASH (Dashboard) option of group System shows a lot of information about the instance.

SDSF MENU 3.1 LOCAL VS01
COMMAND INPUT ==> █

NP	NAME	Description	Group	Status
	SYM	System symbols	System	
	SYS	System information	System	
	ENQ	Enqueues	System	
	ENQC	Enqueue contention	System	
	DYNX	Dynamic exits	System	
	CK	Health checker	System	
	LNK	Link list data sets	System	
	LPA	Link pack data sets	System	
	APF	APF data sets	System	
	PAG	Page data sets	System	
	PARM	Parmlib data sets	System	
	SSI	Subsystem information	System	
	GT	Generic tracker	System	
	LLS	Link list sets	System	
	SYSP	System parameters	System	
	LPAR	Logical partitions	System	
	SMFO	SMF options	System	
	SMFD	SMF data sets	System	
	SMFS	SMF subsystems	System	
	DASH	Dashboard	System	
	ELOG	Event log	System	

SDSF SYSTEM DASHBOARD

SDSF DASHBOARD VS01		LOCAL	LINE 1-36 (43)	
COMMAND INPUT ==>			SCROLL ==> CSR	
Attribute	Value		Metric	Measure

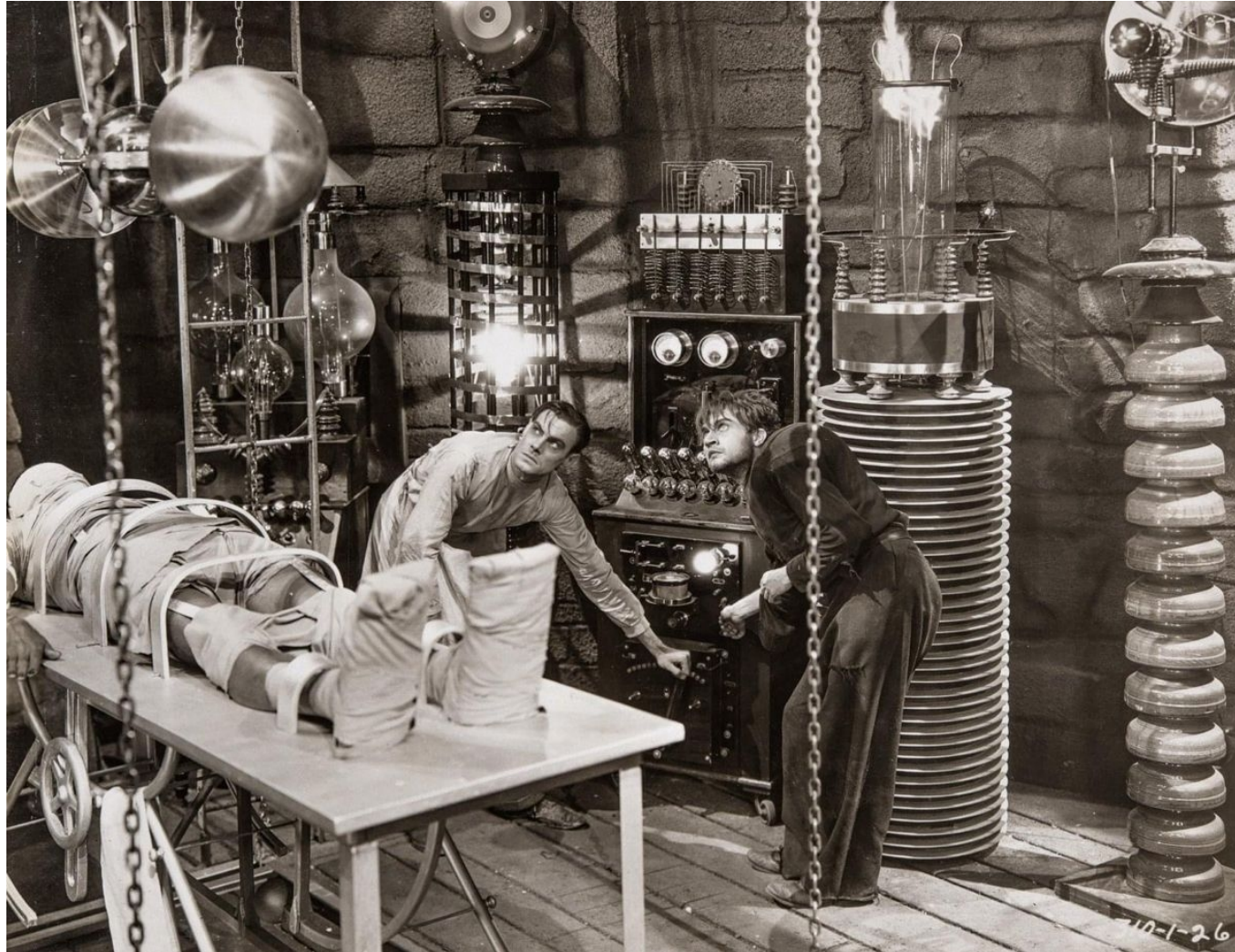
Sysplex	LOCAL		CPU%	13
Sysname	VS01		zIIP%	0
Syslevel	z/OS 03.01.00	HBB77E0	Spool%	0
Sysclone	01		SIO	0
SMFID	VS01		Aux%	0
JESName	JES2		RealAFC	2327986
JESNode	N1		Real%	35
JESMember	VS01		PageRate	0
LPAR	VS01		SysMSU	200
IPLVolser	Z31VS1		AvgMSU	0
IPLDate	2026/06/16-05:46:02		MaxASID	2045
IEASYS	(00)		FreeASID	1947
IEASYM	(00,K2)		BadASID	2
CVTVERID			TSO	1
HardwareName			Job	0
CPCNode	.	.	WTOR	1
Userid	IBMUSER		HVComm%	16
ASID	0065		ECSA%	9
LogonProc	DB13PROC		ESQA%	10
Terminal	TCP00015		CSA%	8
JobID	TSU00178		SQA%	13

Here's a subset of the information displayed on the Dashboard panel. You can see some of the same information as we saw on the SYS panel, plus more detail.

We can see the exact version of the OS we're running, down to the patch level.

We also see some of the information we encountered when exploring the IPL process, like the Nucleus volume serial number and the IEASYS/IEASYM suffixes.

LAB Z/OS 25: SDSF SYSTEM STATUS PANELS



Z/OS SUBSYSTEMS

SDSF MENU 3.1 LOCAL VS01

COMMAND INPUT ==> █

NP	NAME	Description	Group	Status
	JG	Job groups	JES	
	SYM	System symbols	System	
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	SR	System requests	Log	
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	DYNX	Dynamic exits	System	
	AS	Address space memory	Jobs	
	INIT	Initiators	JES	
	PR	Printers	JES	
	PUN	Punches	JES	
	RDR	Readers	JES	
	LINE	Lines	Network	
	NODE	Nodes	Network	
	SO	Spool offload	JES	
	SP	Spool volumes	JES	
	NS	Network servers	Network	
	NC	Network connections	Network	
	RM	Resource monitor	JES	
	CK	Health checker	System	
	LNK	Link list data sets	System	
	LPA	Link pack data sets	System	

Of course, we aren't looking at SDSF panels just for the sake of looking at panels, however enjoyable that may be. We're getting familiar with the terrain, the lay o' the land, the moving parts of the machine, and other clever metaphors like those.

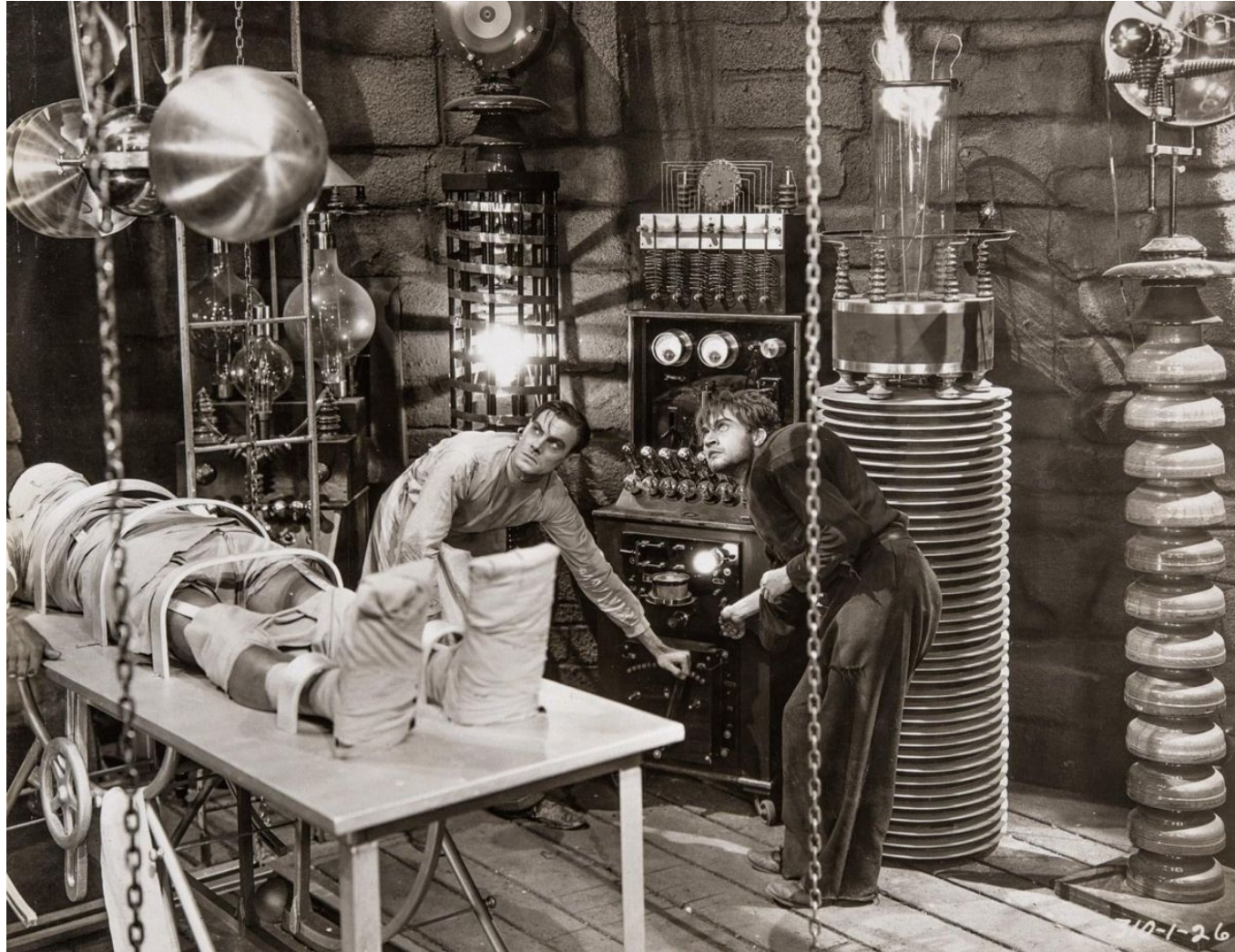
One of the pieces of information you discovered in the last lab was a list of the subsystems currently defined for the z/OS instance. One of them is the Storage Management Subsystem, or SMS.

"Storage" is one of those overloaded terms. On z/OS, the word "storage" means "main memory." It also means peripheral storage – where data sets and files reside.

SMS pertains to the latter definition of "storage." It's the z/OS subsystem responsible for managing (virtual) DASD devices and other traditional mainframe peripheral devices like tape drives, printers, card readers, and card punches.

In the next lab you'll practice locating and reading IBM documentation and correlating what you learn there with the information found on SDSF panels pertaining to SMS.

LAB Z/OS 26: LEARN ABOUT SMS



TYPES OF WORKLOADS



Different computing workloads have different characteristics. Some common types of workloads include:

- Batch – processing a (possibly very large) batch of input data in a single run (e.g., ACH, monthly payroll).
- Transactional – processing one transaction at a time from clients, typically with a human user waiting for the response (e.g., funds transfer request, online purchase).
- OLTP – transactional, but processing very short requests and possibly very high-volume requests, typically from automated clients or as a single step in a transaction (e.g., stock trading floor, credit authorization service).

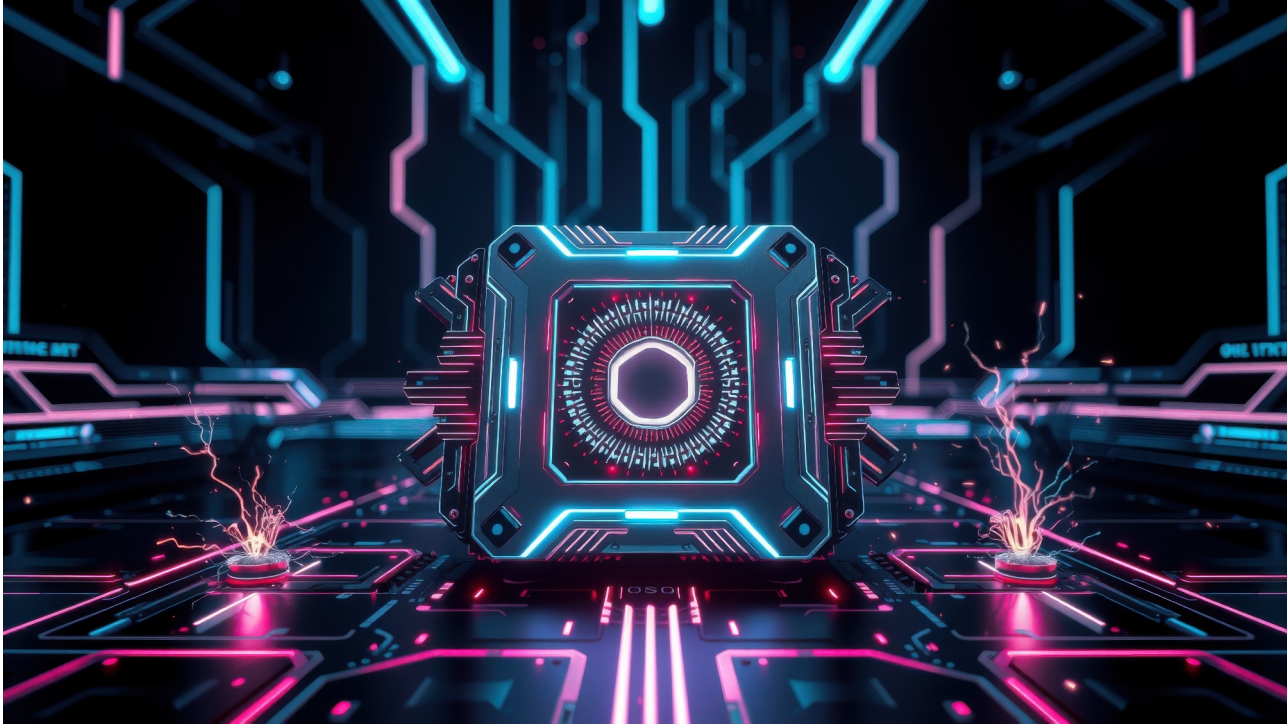
WORKLOAD I/O CHARACTERISTICS

Workloads may differ in their I/O patterns, as well. Some may involve heavy read activity but light write activity, and others the opposite. Some may require ACID properties while others can accept eventual consistency.

- Banking applications – ACID properties necessary
- Social messaging (like Twitter) – eventual consistency is acceptable



WORKLOAD COMPUTE CHARACTERISTICS



Some workloads may place heavy demand on compute resources while others don't.

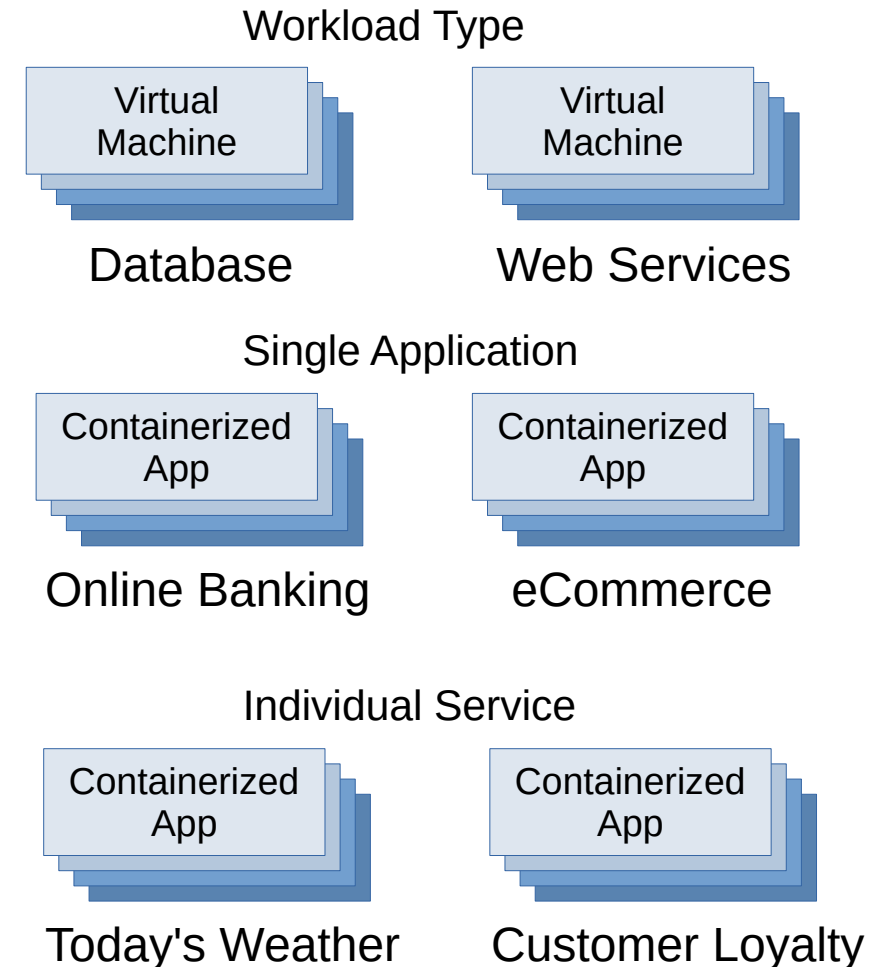
- Large Language Model processing
- Scientific/Engineering applications
- Decision Support or What-if analysis

DISTRIBUTED OR CLOUD INFRASTRUCTURE

From your previous experience, you are probably accustomed to managing different workloads in a distributed or cloud environment. Typically this is handled by defining groups of virtual servers or containerized apps dedicated to

- a particular type of workload, like serving HTTP requests or managing database access;
- a specific application, like Online Banking or eCommerce; or
- an individual microservice, like "Get today's weather report" or "Retrieve customer loyalty points for customer id xxxxx."

The servers in each group are configured appropriately to handle the characteristics of the workload they support.



Z/OS WORKLOAD MANAGEMENT

The IBM Z (or its ancient ancestor, anyway) was born in the era of centralized computing, when a single computer had to be All Things to All Users. Today, a Z can host a multitude of applications simultaneously.

Each operating system instance running in the Sysplex can be thought of as a server in a network, whether it's running in its own LPAR or as a guest under one of the hypervisors, z/VM or KVM.

A single instance of z/OS can host many applications that have quite different operational characteristics.

For instance, our lab environment is a single z/OS LPAR, and it includes both batch processes and a CICS environment for interactive applications, as well as TSO, and it supports VSAM, DB2, and IMS data access.

All those workload types have to be managed within the scope of the single z/OS instance. The z/OS subsystem that handles this the Workload Manager, or WLM.



Z/OS WORKLOAD MANAGEMENT

COMMAND INPUT ==>

NP	NAME	MaxPer	WorkLoad	Goal
	HOTBATCH	1	PRDBAT	Execution velocity of 50
	KILLIT	1	STC	Discretionary
	NEWWORK	2	NEWWORK	80% complete within 00:00:00.500
	NEWWORK	2	NEWWORK	Execution velocity of 20
	NEWWORKV	1	NEWWORK	Execution velocity of 20
	ONLPRD	1	ONLINE	Execution velocity of 50
	ONLTST	1	ONLINE	Execution velocity of 10
	PRDBATHI	1	PRDBAT	Execution velocity of 40
	PRDBATLO	1	PRDBAT	Discretionary
	PRDBATMD	1	PRDBAT	Execution velocity of 5
	SERVERS	1	ONLINE	Execution velocity of 70

WLM is part of RMF, Resource Management Facility, and the documentation for WLM is found under the heading of RMF.

To use WLM, we create a *service definition* for each type of workload we want to manage. The service definition consists of *service classes*, which are categories of work meaningful in our organization, grouped into *workloads* (also called *workload groups*).

We define *goals* (sometimes called *expectations*) for the performance of each service class, where "performance" can mean "response time" or "throughput" or

something along those lines, depending on what's useful to us in our environment.

WLM measures the performance of service classes and tries to achieve the defined goals by adjusting resource allocations and priorities across all the running processes on the system.

WLM can do this without further configuration or action, but it can also cooperate with applications that are designed to call its APIs. These include IBM products like DB2, CICS and others. Documentation for WLM is scattered throughout the documentation for other IBM products and components of z/OS.

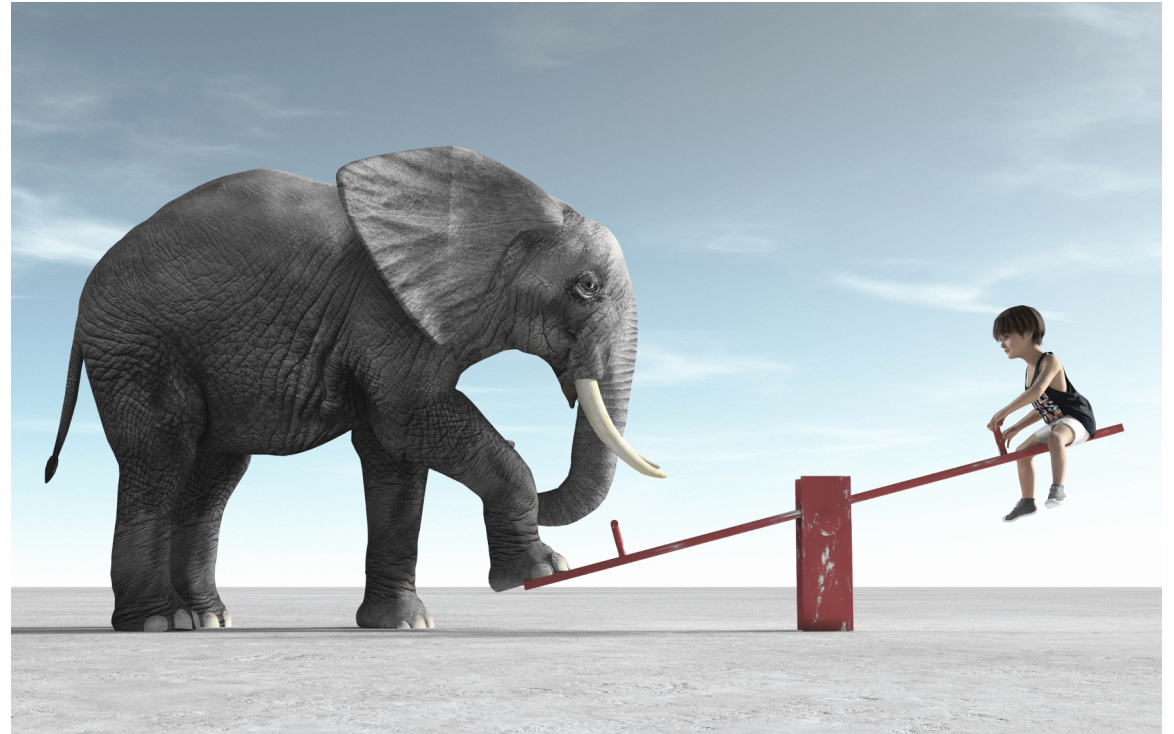
A subset of information from the SDSF WLM Service Classes panel is shown here. You can see what typical goals look like.

Z/OS WLM AND NETWORKING

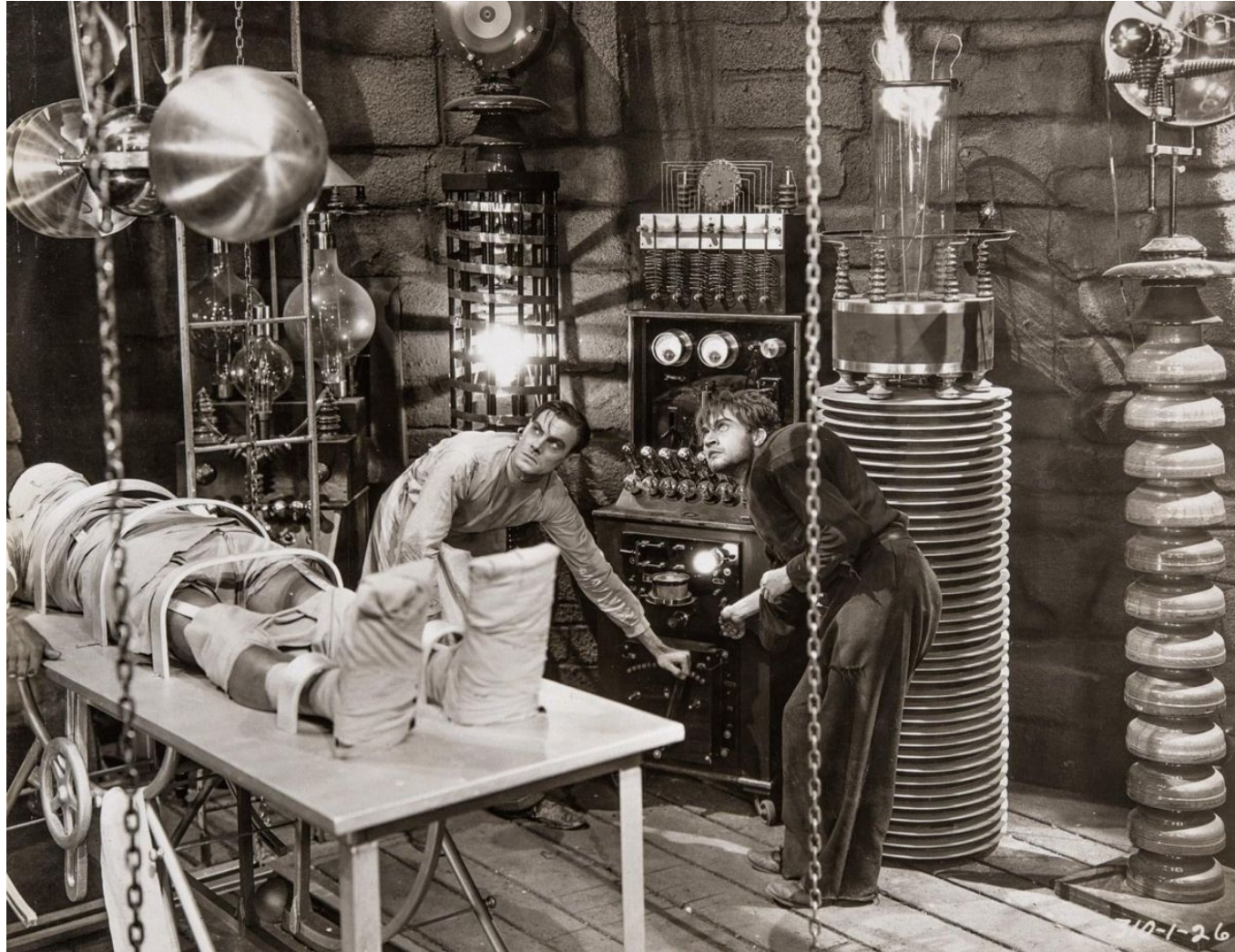
Originally, WLM sought to achieve maximum throughput across all the workloads on the system while honoring high-priority work. It did so (and still does) by dynamically allocating system resources to various running tasks.

As the Z evolved from a single, centralized computer to a sort of network-in-a-box, another workload management function became necessary: Load balancing.

Working in conjunction with the z/OS TCP/IP stack, WLM routes inbound requests to the appropriate LPARs to balance the load and maintain high throughput.



LAB Z/OS 27: LEARN ABOUT WLM



WHAT YOU KNOW AND WHAT YOU CAN DO

- You know how to find information about the z/OS system you're signed on to using SDSF panels.
- You know a bit about SMF
- You know a bit about WLM
- You're getting better at finding IBM documentation and related material online
- You're beginning to cultivate a taste for alphabet soup.

WHAT'S NEXT

- z/OS System Security and RACF